

Matrix functions and stability

Multiplication overhead of cubic sign compositions

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because optimal cubic compositions must be compared with unrestricted evaluation programs; community impact comes from reusable matrix-sign iterations.

Context and notation

For $m \in \mathbb{N}_0$ and $0 < \delta < 1$, put

$$I_\delta = [-1, -\delta] \cup [\delta, 1].$$

Let $\mathcal{P}_m$ consist of real polynomials computed from $1, x$ by straight-line programs using at most m non-scalar multiplications; real linear combinations cost nothing. Define

$$E_m(\delta) = \inf_{p \in \mathcal{P}_m} \max_{x \in I_\delta} |p(x) - \text{sign}(x)|.$$

The arithmetic model concerns a single polynomial identity valid for matrices of every size.

Problem statement

Define

$$C_T(\delta) = \inf_{a_t, b_t \in \mathbb{R}} \max_{x \in I_\delta} |(q_T \circ \dots \circ q_1)(x) - \text{sign}(x)|, \quad q_t(x) = a_t x + b_t x^3.$$

Determine the asymptotic dependence on m, δ of

$$T_{\min}(m, \delta) = \inf\{T \in \mathbb{N}_0 : C_T(\delta) \leq E_m(\delta)\},$$

where the empty composition is x and $\inf \emptyset = +\infty$. Each stage costs at most two matrix products.

References and status evidence

Amsel et al., [Simons workshop report](#), §6.3, Problem 6.5. Rubensson, Jarlebring and Lorentzon, [degree-eight recursive expansion](#), §7, provides a June 2026 follow-up discussion; it does not settle this comparison.

Scope

[MF-01](#) asks for an optimum value over general evaluation programs. This problem measures the price of a particular, reusable composition architecture; the two tasks are related but have different requested outputs.

Audit — 2026-09-10

Rechecked [Problem 6.5](#) and the [degree-eight paper's concluding discussion](#). The optimal composition comparison remains posed. Searches for cubic and recursive sign-expansion improvements found no resolution of this exact multiplication-overhead target.